TIMOTHY YOUNG

Senior Backend Developer

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linkedin.com/in/tim-pro/ Austin, TX, United States

EDUCATION

Bachelor's Degree, Computer Science

St. Edward's University

2009 - 2013 United States

SKILLS

JavaScript · TypeScript · Node.js · Express · NestJS · Python · Django · Flask · FastAPI · React · Next.js · AWS(Lambda · EC2 · S3 · ECS · Chatbot) · GCP · Firebase · MongoDB · MySQL · PostgreSQL · DynamoDB · Redis · OAuth · JWT · RabbitMQ · WebRTC · WebSockets · Socket.io · Jenkins · Remix · tRPC · GraphQL · Selenium · Restful APIs · Google Map API · Cypress · Jest · LangChain · Docker · Kubernetes · Github · GitLab

KEY ACHIEVEMENTS

Microservices Migration

Reduced system downtime by 25% by migrating from monolithic to microservices using Node.js and Python.

Real-Time Data Processing

Reduced data latency by 40% by implementing WebSockets and Kafka for real-time transaction processing.

CI/CD Automation

Reduced deployment times by 60% by automating backend deployment pipelines with GitHub Actions and Docker.

LANGUAGES

English Proficient

SUMMARY

Senior Backend Developer specializing in Node.js (NestJS, Fastify) and Python (Django, FastAPI) with expertise in building scalable and high-performance backend systems. Skilled in designing microservices architectures, leveraging event-driven systems (Kafka, RabbitMQ, Redis) for efficient data flow. Experienced in real-time data processing using WebSockets and LiveKit. Proficient in building and optimizing GraphQL (Apollo, Graphene) and RESTful APIs with tRPC. Strong knowledge of PostgreSQL, Redis, and NoSQL databases. Familiar with cloud infrastructure (AWS, GCP), Kubernetes, and Terraform, with hands-on experience in serverless computing using AWS Lambda and Cloudflare Workers. Skilled in CI/CD automation using GitHub Actions and Docker. Passionate about API security (OAuth2, WebAuthn) and backend optimization for creating reliable, resilient, and scalable systems.

EMPLOYMENT HISTORY

Staff Python Engineer — 09/2024 - Present

CivicLift — Litchfield, CT

- Engineered a highly automated migration tool to extract and transform municipal website data, reducing manual effort and optimizing onboarding processes using Python, web scraping, and data engineering techniques.
- Developed custom scrapers for structured and unstructured data extraction, leveraging BeautifulSoup and Selenium for parsing and interacting with dynamic content, while integrating 2Captcha to handle bot detection mechanisms.
- Enhanced scraping efficiency with AI-powered crawling, adapting and extending Firecrawl API to dynamically adjust crawling strategies across diverse municipal platforms.
- Built an ETL pipeline for processing and structuring extracted data, utilizing pandas for transformation, SQLAlchemy for seamless database integration, and Google Sheets API and Airtable API for data validation before platform ingestion.
- Implemented real-time data streaming with WebSocket to enable continuous updates and synchronization between external data sources and the migration pipeline.
- Created a migrating admin interface using React, providing an intuitive UI that interacts with the backend functions.

Senior Backend Developer and Architect — 09/2020 - 08/2024

Fortunes Tech — Liverpool, United Kingdom

- Led the backend modernization of a logistics tracking platform, transitioning from a monolithic system to a microservices-based architecture using Node.js (Express.js) and Python (FastAPI).
- Developed a high-availability real-time tracking system with WebSockets (Socket.IO), Redis Pub/Sub, and Firebase Firestore, ensuring seamless event-driven updates.
- Implemented AI-powered route optimization with Python (Django, TensorFlow) to enhance logistics efficiency and automated decision-making.
- Designed and deployed an event-driven architecture using RabbitMQ and Apache Kafka, enabling asynchronous processing and scalable event distribution.
- Automated infrastructure deployment and orchestration with Terraform, Kubernetes (EKS), and AWS Fargate, ensuring dynamic scaling and high availability.
- Integrated essential third-party services such as Stripe for payments and Google Maps API for real-time location tracking, optimizing platform functionality.
- Developed a fault-tolerant, scalable backend, leveraging Redis caching, asynchronous task processing (Celery), and optimized database queries to ensure performance under heavy load.
- Maintained high system reliability by implementing observability and logging solutions using Grafana and Prometheus.

CERTIFICATION

Certified Python Developer

PCAP

Python Institute Certified Entry-Level Python Programmer

PCEP

AWS Certified Developer

Associate

EMPLOYMENT HISTORY

Backend Developer

02/2015 - 08/2020

Cocolevio

Austin, TX

- Transformed a legacy backend to a microservices architecture using Node.js (NestJS, Express) and RabbitMQ for message-driven communication.
- Leveraged PM2 clustering to improve Node.js application performance and scalability.
- Replaced REST APIs with GraphQL (Apollo Server), optimizing data queries and reducing client-side overhead.
- Enhanced database performance by adding Redis caching and implementing PostgreSQL indexing and read replicas.
- Strengthened security by integrating OAuth2, JWT, and AWS Secrets Manager for key management.
- Enabled real-time communication features via WebRTC and WebSockets in a logistics system.
- Applied Redis and Nginx to build rate limiting and enhance API security.
- Transitioned backend infrastructure to AWS, utilizing Lambda, S3, and DynamoDB for improved scalability.
- Automated deployments with Jenkins and GitHub Actions, streamlining testing, builds, and release management.

Node JS Developer

10/2013 - 12/2014

Praxent

Austin, TX

- Built API Gateway systems using Node.js to support high-performance financial algorithms and transaction handling.
- Implemented event-driven processing to optimize financial data flow using Redis and RabbitMQ for real-time transaction validation.
- Developed algorithms for real-time fraud detection, analyzing patterns and flagging suspicious transactions with minimal delay.
- Designed a dynamic pricing algorithm that adjusts financial product rates in real time based on market trends and user data, powered by Node.js and MySQL.
- Created low-latency trading algorithms for executing high-frequency trades, ensuring efficient execution of financial transactions.
- Integrated data encryption and secure transaction methods, complying with industry standards to protect financial data.
- Streamlined API performance by transitioning from monolithic to microservices architecture.
- Automated testing and deployment pipelines with Jenkins and Docker.